

Report of Calibration
PSC ALSu_1ST_ARTICLE_4-CH-SLOW_HSSS/N 0001
2026-09-23 10:36:24

Calibration Current standard: BNL PSC ATE S/N 001
Calibration Volt standard: HP 3458A-002 S/N 2823A 23647
Calibration Resistance standard: Fluke 742A-1 S/N 1063008
End Header

lab{2}Chan1
Burden resistor = 33.3333

Measuring initial gains and offsets

Itest	dacSP	dcct1	dcct2	dacRB	err
-1.003689	1.005207	-1.013376	-1.014582	1.005370	0.031115
-27.000843	27.004545	-26.827026	-26.831093	27.079170	0.039815

	dacSP	dcct1	dcct2	dacRB
Initial measured offsets:	-0.001434	-0.016771	-0.017867	0.000000
Initial measured gains:	1.000084	0.992941	0.993051	1.002864
Gain corrections:	1.000084	1.007109	1.006997	1.000000

Writing gain and offset corrections for dacSP, dcct1, and dcct2 to PSC

Measuring DAC readback gain and offset

DAC SP	DAC RB
1.000000	1.001428
27.000000	27.078541

Measured offset: -0.001538
Measured gain: 1.002966
Gain correction: 0.997043

Writing gain and offset constants for dacRB to PSC

Verification

Itest	dacSP	dcct1	dcct2	dacRB	err
-1.003705	1.003748	-1.004005	-1.003995	1.003560	0.001155
-27.001111	27.001035	-27.001013	-27.001024	27.000513	0.051801

	dacSP	dcct1	dcct2	dacRB
Final measured offsets:	-0.000047	-0.000315	-0.000304	-0.000175
Final measured gains:	0.999995	0.999985	0.999986	0.999987

	dacSP	dcct1	dcct2	dacRB
Final measured offsets mean:	0.000036	-0.000102	-0.000077	-0.000111
Final measured offsets stdev:	0.000048	0.000148	0.000153	0.000131
Final measured gains mean:	0.999991	0.999989	0.999990	0.999998
Final measured gains stdev:	0.000016	0.000016	0.000015	0.000013

Saving channel 1 calibration constants to qspi

lab{2}Chan2
Burden resistor = 33.3333

Measuring initial gains and offsets

Itest	dacSP	dcct1	dcct2	dacRB	err
-1.003322	1.006509	-1.012174	-1.011059	1.003961	0.024549
-27.000365	27.009073	-26.823298	-26.808178	27.115738	0.051070

	dacSP	dcct1	dcct2	dacRB
Initial measured offsets:	-0.002974	-0.016027	-0.015453	0.000000
Initial measured gains:	1.000212	0.992848	0.992310	1.004200
Gain corrections:	1.000212	1.007203	1.007750	1.000000

Writing gain and offset corrections for dacSP, dcct1, and dcct2 to PSC

Measuring DAC readback gain and offset

DAC SP	DAC RB
1.000000	1.000474
27.000000	27.115240

Measured offset: -0.003940
Measured gain: 1.004414
Gain correction: 0.995605

Writing gain and offset constants for dacRB to PSC

Verification

Itest	dacSP	dcct1	dcct2	dacRB	err
-1.003279	1.003277	-1.003265	-1.003237	1.003873	0.021757
-27.000647	27.000060	-27.000063	-27.000051	26.999832	0.048059

	dacSP	dcct1	dcct2	dacRB
Final measured offsets:	-0.000020	-0.000008	0.000021	0.000628
Final measured gains:	0.999977	0.999978	0.999979	0.999968

	dacSP	dcct1	dcct2	dacRB
Final measured offsets mean:	0.000112	0.000028	0.000024	0.000726
Final measured offsets stdev:	0.000069	0.000080	0.000077	0.000071
Final measured gains mean:	0.999990	0.999988	0.999986	0.999982
Final measured gains stdev:	0.000012	0.000011	0.000009	0.000008

Saving channel 2 calibration constants to qspi

lab{2}Chan3
Burden resistor = 33.3333

Measuring initial gains and offsets

Itest	dacSP	dcct1	dcct2	dacRB	err
-1.002992	1.005450	-1.012213	-1.012362	1.006537	0.010433
-27.000533	26.991986	-26.801834	-26.820883	27.058483	0.042423

	dacSP	dcct1	dcct2	dacRB
Initial measured offsets:	-0.002883	-0.017242	-0.016663	0.000000
Initial measured gains:	0.999577	0.992002	0.992729	1.002517
Gain corrections:	0.999577	1.008062	1.007324	1.000000

Writing gain and offset corrections for dacSP, dcct1, and dcct2 to PSC

Measuring DAC readback gain and offset

DAC SP	DAC RB
1.000000	1.003714
27.000000	27.057556

Measured offset: 0.001643
Measured gain: 1.002071
Gain correction: 0.997933

Writing gain and offset constants for dacRB to PSC

Verification

Itest	dacSP	dcct1	dcct2	dacRB	err
-1.002695	1.002716	-1.002737	-1.002745	1.003506	0.014382
-27.000578	27.000571	-27.000515	-27.000498	27.001724	0.046948

	dacSP	dcct1	dcct2	dacRB
Final measured offsets:	-0.000022	-0.000046	-0.000056	0.000776
Final measured gains:	0.999999	0.999996	0.999995	1.000014

	dacSP	dcct1	dcct2	dacRB
Final measured offsets mean:	-0.000047	-0.000002	-0.000006	0.000697
Final measured offsets stdev:	0.000033	0.000097	0.000089	0.000088
Final measured gains mean:	0.999988	0.999990	0.999990	1.000015
Final measured gains stdev:	0.000011	0.000005	0.000006	0.000005

Saving channel 3 calibration constants to qspi

lab{2}Chan4
Burden resistor = 33.3333

Measuring initial gains and offsets

Itest	dacSP	dcct1	dcct2	dacRB	err
-1.003022	1.000324	-1.011619	-1.011839	1.003774	0.034556
-27.000700	26.998298	-26.809547	-26.823013	27.108133	0.043110

	dacSP	dcct1	dcct2	dacRB
Initial measured offsets:	0.002710	-0.016304	-0.016012	0.000000
Initial measured gains:	1.000011	0.992317	0.992826	1.004092
Gain corrections:	1.000011	1.007743	1.007226	1.000000

Writing gain and offset corrections for dacSP, dcct1, and dcct2 to PSC

Measuring DAC readback gain and offset

DAC SP	DAC RB
1.000000	1.000611
27.000000	27.107376

Measured offset: -0.003495
Measured gain: 1.004106
Gain correction: 0.995910

Writing gain and offset constants for dacRB to PSC

Verification

Itest	dacSP	dcct1	dcct2	dacRB	err
-1.003085	1.002903	-1.003018	-1.003018	1.002826	0.043585
-27.000558	27.000232	-27.000263	-27.000280	26.999580	0.045453

	dacSP	dcct1	dcct2	dacRB
Final measured offsets:	0.000176	0.000058	0.000059	-0.000055
Final measured gains:	0.999994	0.999991	0.999992	0.999978

	dacSP	dcct1	dcct2	dacRB
Final measured offsets mean:	0.000185	0.000073	0.000050	0.000289
Final measured offsets stdev:	0.000192	0.000056	0.000034	0.000321
Final measured gains mean:	0.999998	0.999996	0.999994	0.999988
Final measured gains stdev:	0.000014	0.000008	0.000009	0.000007

Saving channel 4 calibration constants to qspi

Test data reviewed by _____ Date_____